23CSE399

Motion Consistency Evaluation in Unmodified Surveillance Video Data

PROJECT PHASE - 1

Our Team

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Domain:

Image and Video Processing

Problem Statement

Surveillance camera systems generate continuous video streams, but real-world feeds often suffer from hidden issues such as prolonged low motion, frozen frames, or irregular temporal behavior due to environmental or hardware problems. These issues may not trigger traditional object-based alerts yet significantly reduce monitoring reliability.

This project evaluates **motion consistency in unaltered surveillance videos** by analyzing natural temporal motion patterns to detect irregularities and assess **camera feed health and reliability**.

Originality of Work

Existing surveillance research primarily focuses on identifying *what* moves in a scene through object and event detection, whereas this work shifts the focus to *how motion behaves over time*. The proposed approach introduces **motion consistency** as a novel indicator of camera health and reliability, enabling continuous assessment without relying on object detection, manual labeling, or simulated attacks. By operating directly on unmodified, real-world surveillance datasets, this method avoids intrusive system changes. Framing motion consistency as a **temporal reliability metric** for surveillance camera networks, rather than as an activity or anomaly detection task, establishes the originality and research contribution of this work.

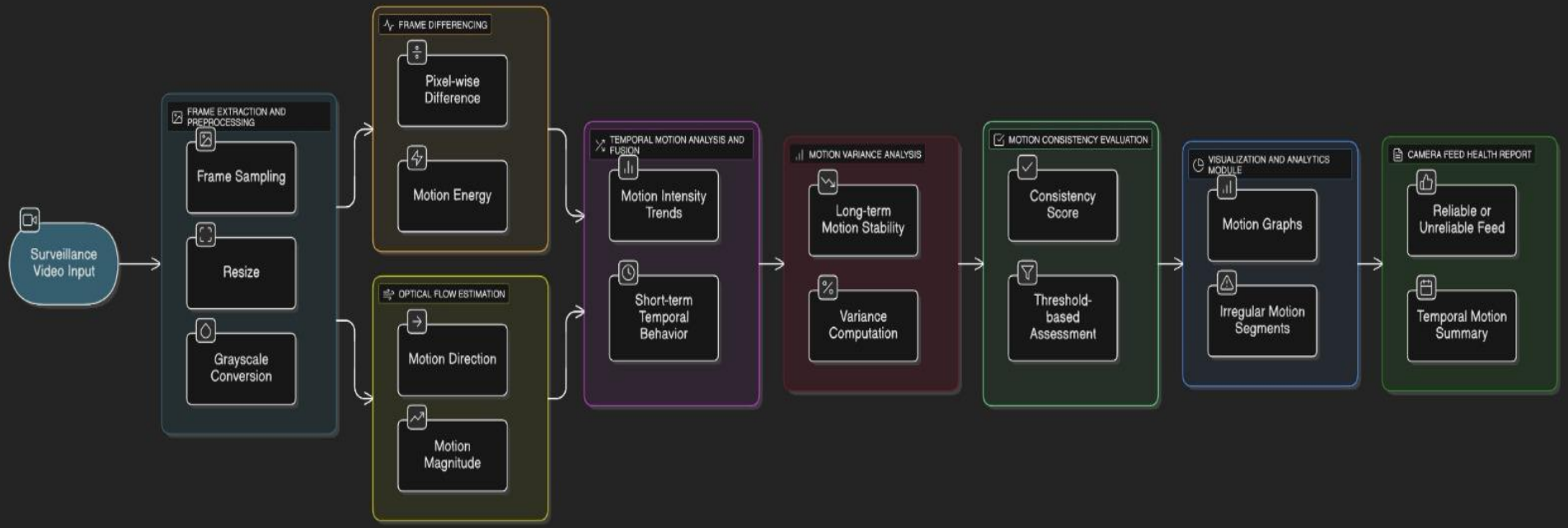
Motivation

Most surveillance systems continuously capture video but lack effective temporal monitoring of camera health, as existing object and event detection methods assume reliable camera operation. In practice, camera feeds can silently degrade due to environmental changes, hardware aging, or network issues, which object-based analytics fail to detect. Prior research shows that motion statistics are closely linked to video quality and feed reliability, indicating that motion behavior reflects the operational usefulness of a camera. Therefore, temporal motion analysis offers an effective way to identify reliability issues early, highlighting a clear research gap in motion-based camera health analytics.

Problem Definition

The system is designed to analyze **natural motion patterns** in surveillance videos and measure **motion consistency over time** to assess camera feed reliability. It aims to detect prolonged low-motion periods, sudden drops in motion, and irregular temporal motion behavior that may indicate feed degradation or malfunction. The approach operates on **unmodified real-world surveillance datasets** and focuses on providing **motion-based indicators** for continuous camera health monitoring, without relying on object detection or artificial attack scenarios.

Architecture Diagram



Methodology

1. **Frame Differencing**
   - Computes pixel-level differences between consecutive frames.
   - Captures instantaneous motion intensity.
   - Simple and efficient for continuous surveillance monitoring.

2. **Optical Flow**
   - Computes direction and magnitude of motion across frames.
   - Captures smooth and gradual movement patterns.
   - Suitable for pedestrian and vehicle motion analysis

3. **Temporal Motion Intensity Analysis**
   Aggregates motion values over sliding time windows.
   Produces temporal motion intensity curves.
   Helps identify prolonged low-motion or irregular temporal behavior.

4. **Variance-Based Consistency Analysis**
   Computes statistical variance of motion over time.
   High variance indicates unstable or irregular camera feed.
   Low variance indicates consistent and reliable motion behavior.

Dataset Description

ShanghaiTech Campus Dataset

- Real-world surveillance video dataset
- Contains both **normal and abnormal activity sequences**
- Well suited for **long-term temporal motion analysis**

UCSD Pedestrian Dataset (Ped1 & Ped2)

- Controlled pedestrian surveillance environment
- Widely used **standard benchmark dataset**
- Suitable for evaluating surveillance reliability models

Avenue Dataset

- Outdoor surveillance scenes
- Captures **natural motion variations**
- Useful for analyzing real-world motion behavior

VIRAT Video Dataset

- Large-scale, realistic surveillance footage
- Complex scenes with diverse activities
- Widely used in **security and video analytics research**

Literature survey

Paper Title	Authors / Year	Problem Focus	Essence
STAD-AI: Spatio-Temporal Anomaly Detection	2025	Anomaly detection	Uses spatio-temporal motion features to detect abnormal events in surveillance videos
Vision Transformer Embedded VAD	2025	Deep anomaly detection	Learns long-range spatial & temporal motion patterns using transformers
AnomLite: Efficient Video Anomaly Detection	2025	Lightweight VAD	Efficient spatio-temporal motion modeling with temporal learning
MG-KG: Unsupervised VAD	2025	Motion-guided anomaly detection	Uses motion guidance and temporal relations for unsupervised anomaly detection
Fire-Swin: Hybrid Surveillance Model	2025	Event detection	Transformer-based long-range temporal attention for surveillance events

Paper Title	Authors / Year	Problem Focus	Essence
Video Prediction via Temporal Aggregation	2025	Temporal coherence	Models motion continuity using frame prediction and temporal propagation
Quality-Aware Video Anomaly Detection	2024	Video quality & motion	Shows motion statistics reflect video quality and stability
Temporal Motion Descriptor for Video Analysis	2025	Motion descriptors	Proposes temporal motion descriptors to characterize video behavior
Temporal Coherence in Surveillance Videos	2024	Temporal stability	Analyzes temporal coherence as an indicator of video reliability
Motion-Based Consistency Metrics	2025	Motion consistency	Defines metrics to quantify motion consistency in surveillance streams
Real-Time Temporal Consistency Evaluation	2025	Feed consistency	Evaluates temporal consistency of surveillance video streams

Comparison with Existing Approaches

Approach	Focus	Require Labels	Modifies Data	Camera Health
Object Detection	Objects	Yes	No	❌
Event Detection	Events	Yes	No	❌
Anomaly Detection	Activities	Often	Sometimes	❌
Proposed Method	**Motion Consistency**	**No**	**No**	✅

Justification of our work

The proposed method is justified by its use of **real, unmodified surveillance datasets**, ensuring practical applicability. By focusing on **temporal motion behavior**, the approach provides direct **camera feed health indicators** rather than activity-level interpretations. It avoids ethical and legal concerns associated with simulated attack scenarios or data manipulation, while establishing a **strong foundation for future machine learning–based extensions**.

Expected Outcomes

- Quantitative **motion consistency scores** for surveillance video streams
- Detection of:
  - Prolonged **low-motion segments**
  - **Irregular temporal motion periods**
- **Temporal motion intensity graphs** for visual analysis
- **Motion-based camera feed health indicators** for reliability assessment

Application Areas

- Campus and institutional surveillance
- Smart city monitoring systems
- Transportation hubs (airports, metro stations, bus terminals)
- Industrial and warehouse security environments

Thank you!